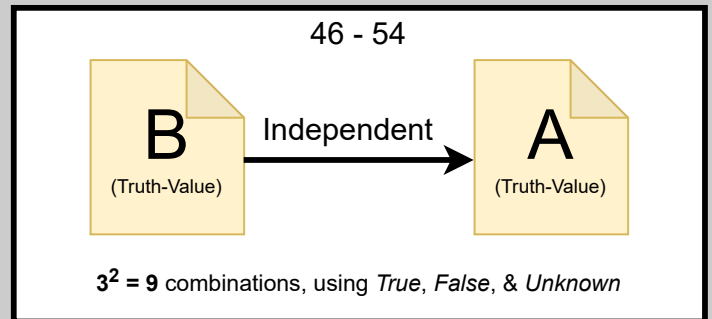
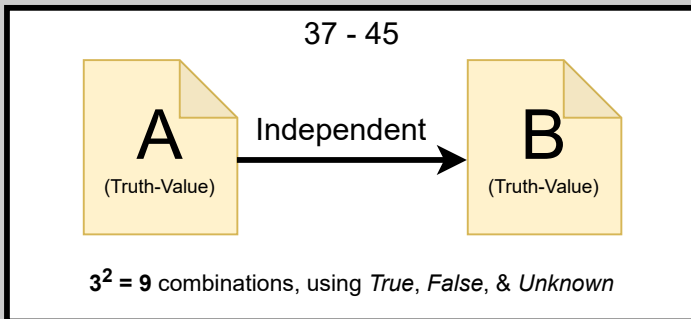
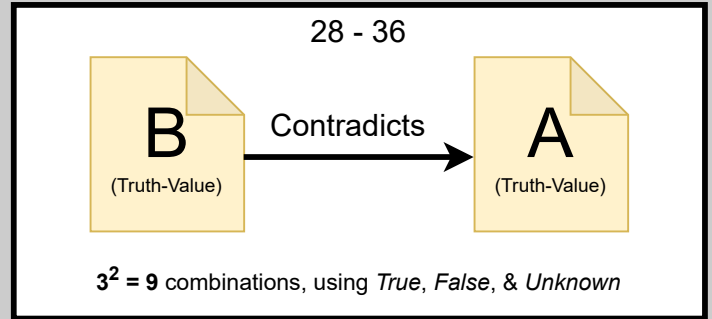
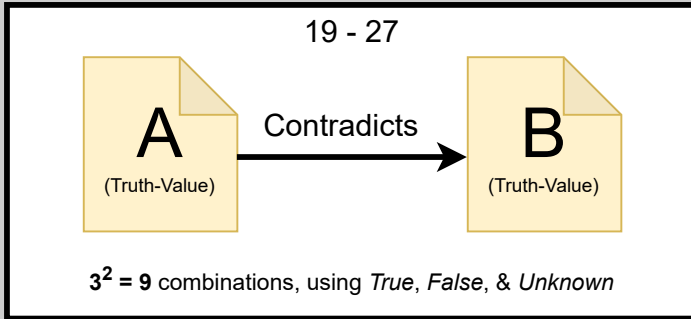
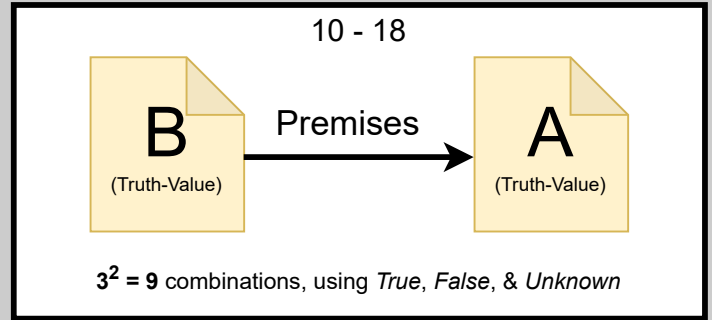
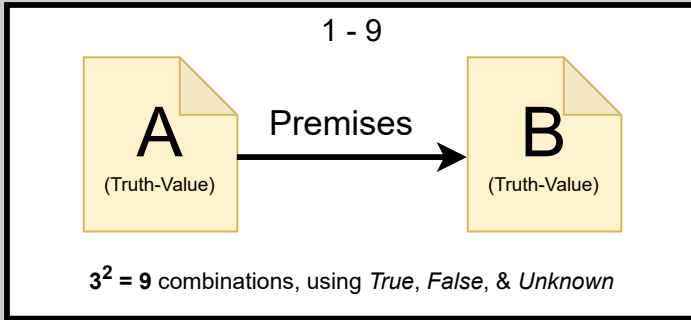


$$N = 2$$



$A_n = 3^n \sum_{p=1}^{n-1} \binom{n}{p} 3^{p(n-p)}$	3^n	n statements, 3 possible truth-value arrangements (<i>true</i> , <i>false</i> , and <i>unknown</i>)*
	$\sum_{p=1}^{n-1}$	count the number of ways n statements can be arranged into premises and conclusions
	$\binom{n}{p}$	for p statements, every possible arrangement of p premises and $n - p$ conclusions*
	$3^{p(n-p)}$	p premises, $n - p$ conclusions, 3 possible connection arrangements (<i>premises</i> , <i>contradicts</i> , <i>independent</i>)

$UpperBound_n = 3^n 2^n 3^{n^2/4}$	3^n	n statements, 3 possible truth-value arrangements (<i>true</i> , <i>false</i> , and <i>unknown</i>)
	2^n	upper bound to pick a subset of n statements to serve as premises
	$3^{n^2/4}$	maximum number of ways to arrange connections, when the premises and conclusions are both $n / 2^{**}$

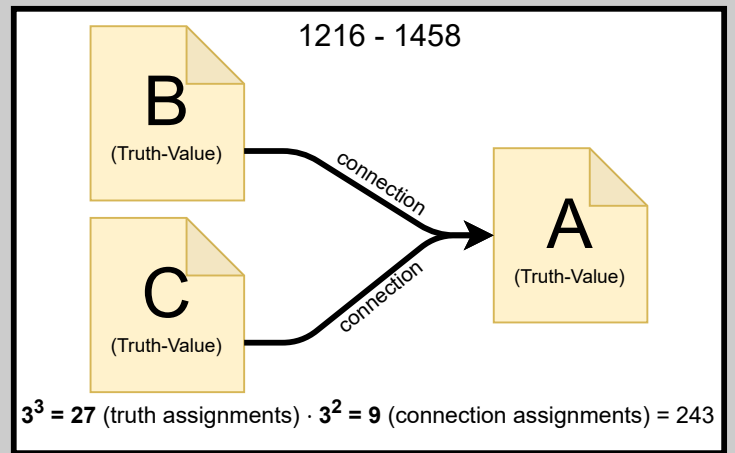
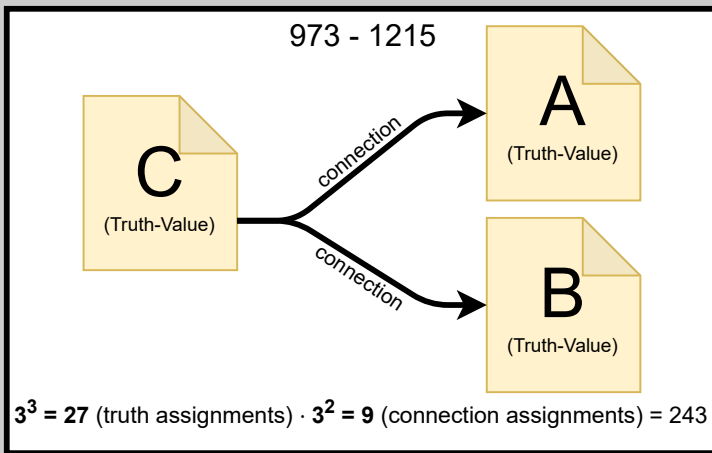
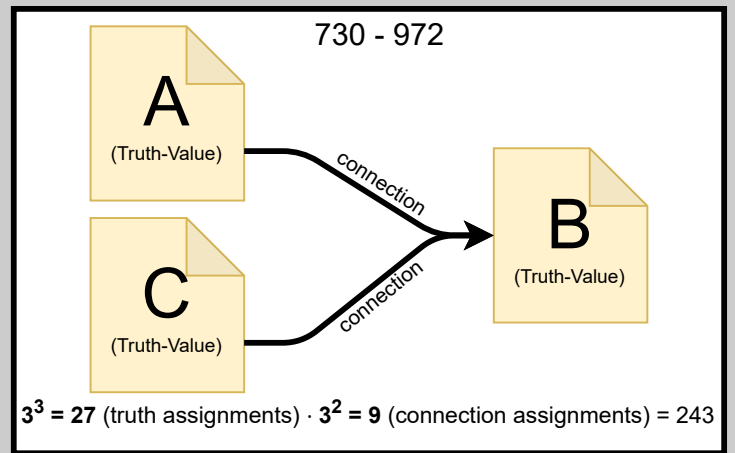
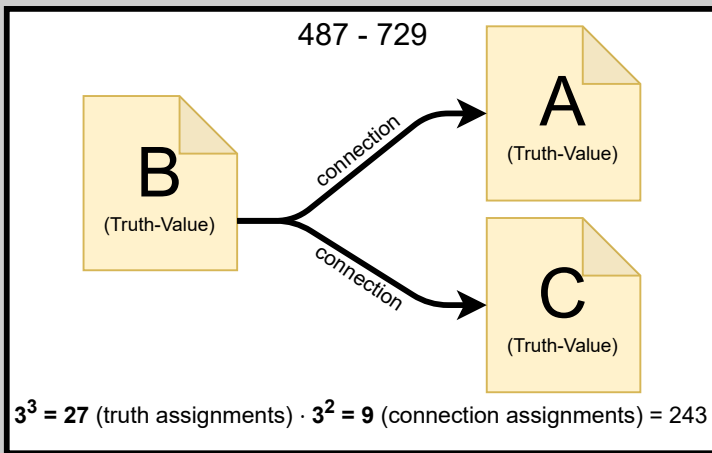
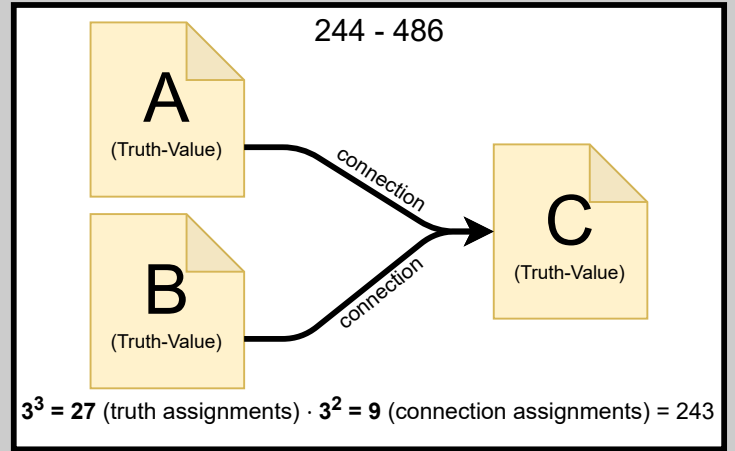
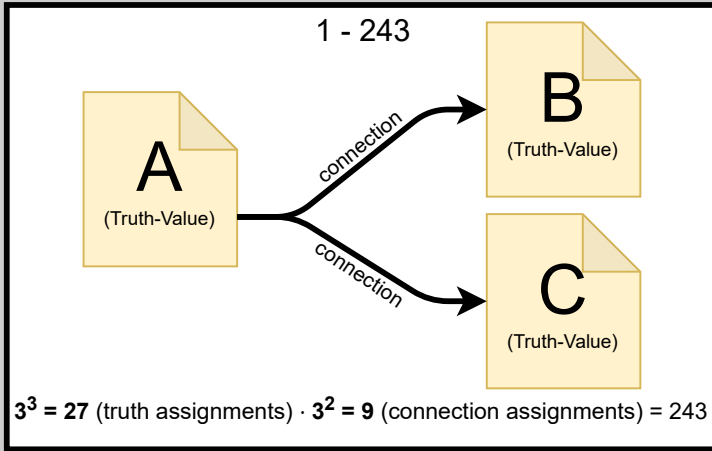
$$A_2 = 3^2 \sum_{p=1}^1 \binom{2}{p} 3^{p(2-p)} = 3^2 \binom{2}{1} 3^1 = 9 * 2 * 3 = 54$$

$$UpperBound_2 = 3^2 2^2 3^{2^2/4} = 9 * 3 * 4 = 108$$

* Semantically equivalent statements, such as "A is independent of B" and "B is independent of A", are considered distinct *

** $(n/2)(n/2) = n^2 / 4$; note that super-exponentials grow much faster than factorials: $2^n < n! < n^n < (k^n)^n = k^{n^2} \propto 3^{n^2/4}^{**}$

$$N = 3$$



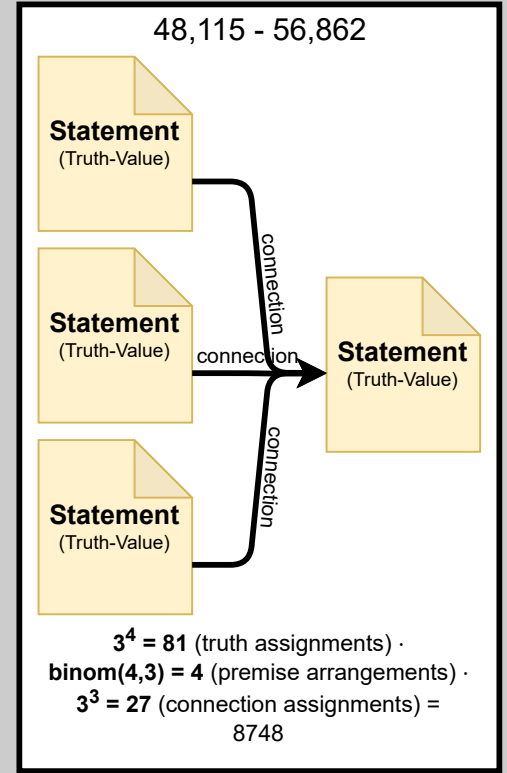
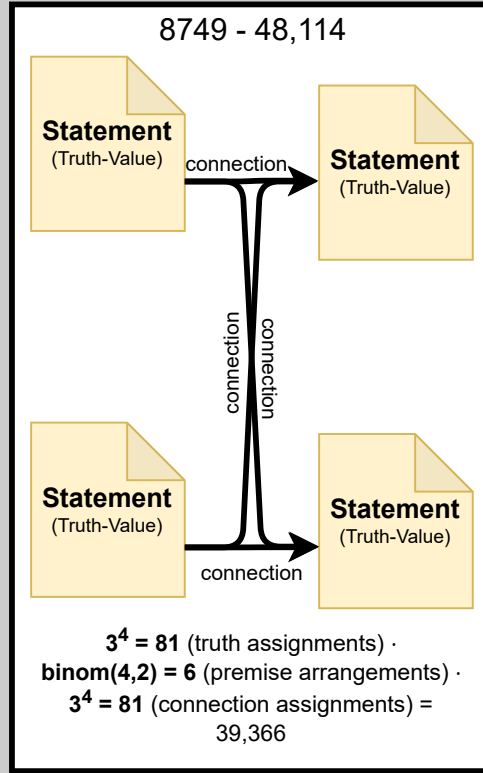
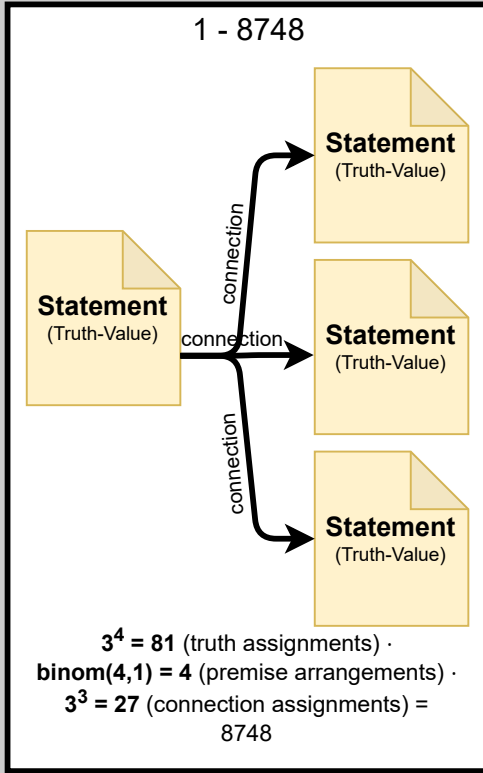
$$A_n = 3^n \sum_{p=1}^{n-1} \binom{n}{p} 3^{p(n-p)}$$

$$UpperBound_n = 3^n 2^n 3^{n^2/4}$$

$$A_3 = 3^3 \sum_{p=1}^2 \binom{3}{p} 3^{p(3-p)} = 3^3 \left(\binom{3}{1} 3^2 + \binom{3}{2} 3^2 \right) = 27(3 * 9 + 3 * 9) = 1458$$

$$UpperBound_3 = 3^3 2^3 3^{3^2/4} = 27 * 8 * 11.8447 \approx 2559$$

$$N = 4+$$



$$A_4 = 3^4 \sum_{p=1}^3 \binom{4}{p} 3^{p(4-p)} = 3^4 \left(\binom{4}{1} 4^3 + \binom{4}{2} 4^4 + \binom{4}{3} 4^3 \right)$$

$$= 81(4 * 27 + 6 * 81 + 4 * 27) = 56,862$$

$$UpperBound_3 = 3^4 2^4 3^{4^2/4} = 81 * 16 * 81 = 104,976$$

$$A_n = 3^n \sum_{p=1}^{n-1} \binom{n}{p} 3^{p(n-p)}$$

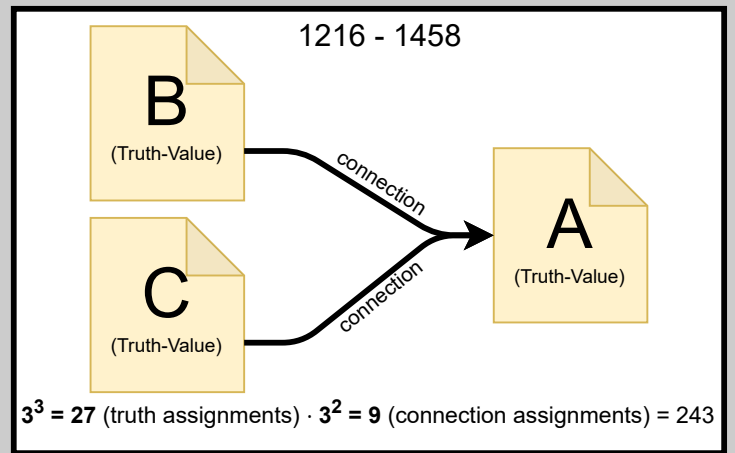
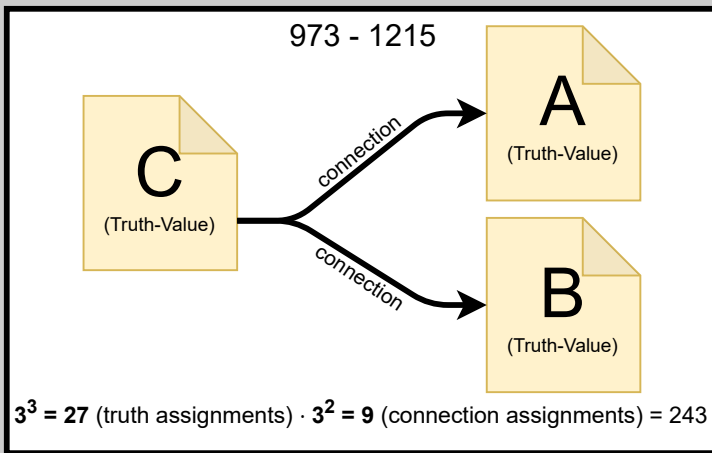
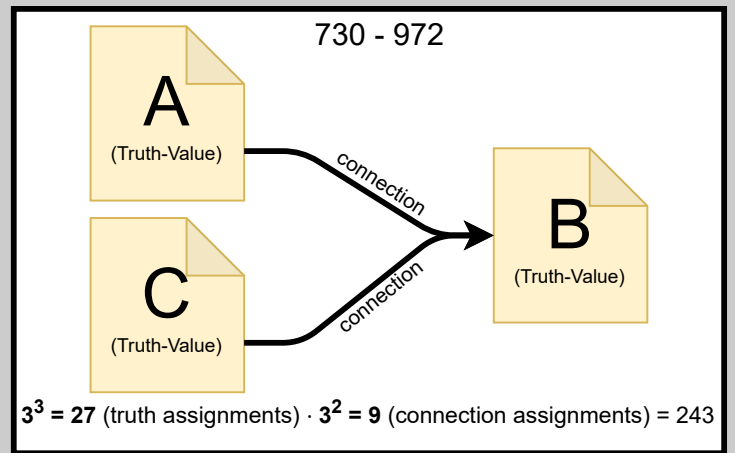
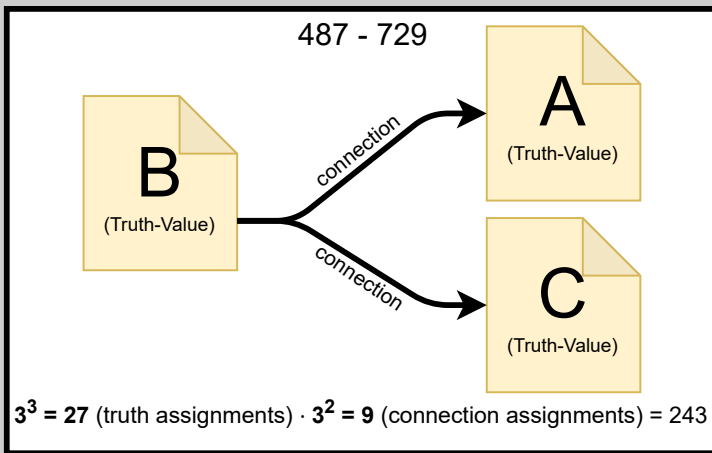
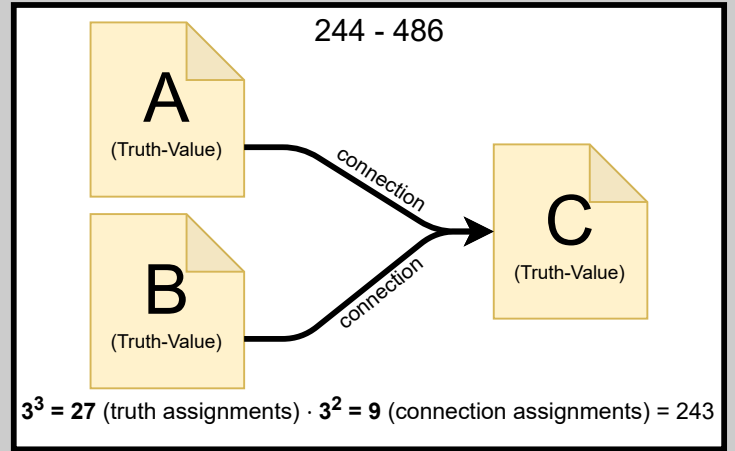
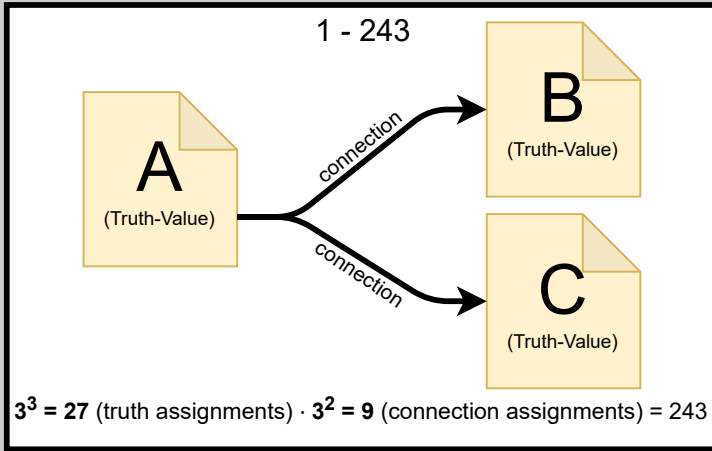
$$UpperBound_n = 3^n 2^n 3^{n^2/4}$$

Note: the number of panels above corresponds to the summation, in which the number of premises in each panel ranges from 1 to $n-1$ (this intuition can be carried over to all other n), with the worst-case middle panel ($n/2$) being used to get the upper bounds

$$\begin{aligned} A_1 &= 1 \\ A_2 &= 54 \\ A_3 &= 1458 \\ A_4 &= 56,862 \\ A_5 &= 3,739,770 \\ A_6 &= 432,592,974 \\ A_7 &= 86,804,510,058 \\ A_8 &= 30,509,593,091,262 \\ A_9 &= 18,582,736,106,826,810 \\ A_{10} &= 19,760,836,085,085,811,374 \end{aligned}$$

$$\begin{aligned} U_1 &\approx 8 \\ U_2 &= 108 \\ U_3 &\approx 2558 \\ U_4 &= 104,976 \\ U_5 &\approx 7,460,434 \\ U_6 &= 918,330,048 \\ U_7 &\approx 195,791,630,461 \\ U_8 &= 72,301,961,339,136 \\ U_9 &\approx 46,245,199,948,311,283 \\ U_{10} &= 51,232,302,181,375,699,968 \end{aligned}$$

$$N = 3$$



$$A_n = 3^n \sum_{p=1}^{n-1} \binom{n}{p} 3^{p(n-p)}$$

$$UpperBound_n = 3^n 2^n 3^{n^2/4}$$

$$A_3 = 3^3 \sum_{p=1}^2 \binom{3}{p} 3^{p(3-p)} = 3^3 \left(\binom{3}{1} 3^2 + \binom{3}{2} 3^2 \right) = 27(3 * 9 + 3 * 9) = 1458$$

$$UpperBound_3 = 3^3 2^3 3^{3^2/4} = 27 * 8 * 11.8447 \approx 2559$$